

LAB EXPERIMENT
ITA0506-COMPUTER VISION

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EXPNO: 36.

Recognise Watch Using OpenCV

Aim:

To recognize a watch from the given image using OpenCV.

Algorithm

1. Import OpenCV.
2. Read the image WATCH.PNG.
3. Convert the image into grayscale.
4. Apply Gaussian blur to reduce noise.
5. Detect the circular watch region using Hough Circle Transform.
6. Draw a circle around the detected watch.
7. Display and save the result.

CODE:

```
EXP-36.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...
File Edit Format Run Options Window Help

import cv2

# Read watch image
img = cv2.imread("WATCH.PNG")

if img is None:
    print("Image not found!")
    print("Make sure the file name is WATCH.PNG")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Reduce noise
gray = cv2.GaussianBlur(gray, (9, 9), 2)

# Detect circles
circles = cv2.HoughCircles(
    gray,
    cv2.HOUGH_GRADIENT,
    dp=1,
    minDist=50,
    param1=100,
    param2=35,
    minRadius=20,
    maxRadius=100
)

# Check if circle is detected
if circles is not None:
    circles = circles.astype(int)

    # Select the first detected circle
    x, y, r = circles[0][0]

    # Draw circle
    cv2.circle(img, (x, y), r, (0, 255, 0), 3)

    # Mark center
```

Ln: 8 Col: 43

OUTPUT :

```
*IDLE Shell 3.13.3*
File Edit Shell Debug Options Window Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-35.py
Black Hat operation completed successfully!

= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-36.py
Watch not detected.

= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-36.py
Watch detected successfully!
```



The image shows a screenshot of a Python IDLE Shell window. The shell displays the output of three script executions. The first script (EXP-35.py) successfully completed a 'Black Hat operation'. The second script (EXP-36.py) failed to detect a watch. The third script (EXP-36.py) successfully detected a watch. Below the shell output, a small window titled 'Watch Recognition' is shown. This window displays an image of a watch with a green circular bounding box around it and the word 'Watch' in green text above it.

Result

The circular region corresponding to the watch is detected and marked using OpenCV.